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The Given–Unknown–Equation–Substitute–Solve (GUESS) Principle for Quantitative STEM Problem Solving: A Mixed-Methods Evaluation Across Four Undergraduate Disciplines

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Abstract

Background: Students often possess isolated formulas yet struggle to represent quantitative problems, select appropriate equations and transfer a workable solution routine across STEM disciplines. Objective: This study evaluated the Given–Unknown–Equation–Substitute–Solve (GUESS) principle as a five-step scaffold for well-structured calculation problems. Methods: A sequential explanatory mixed-methods design was reported across General Chemistry I ($n=78$), Calculus I ($n=65$), Engineering Mechanics ($n=42$) and Introduction to Physics ($n=56$) at the University of Abuja during 2023–2024 ($N=241$). Students were assigned within courses to GUESS or traditional instruction. Pre/post problem-solving tests, structured error analysis, student interviews ($n=40$), classroom observations, instructor journals and think-aloud protocols ($n=20$) were used. Results: Reported pre–post gains were larger in GUESS groups in every course (14.7–18.2 points) than in controls (6.2–8.6 points). The reported two-way ANOVA showed a main effect of teaching approach, $F(1,233)=42.68$, $p<.001$, partial $\eta^2=.155$, with no significant discipline effect or approach×discipline interaction. Equation-selection errors were 16% in the GUESS group versus 36% in controls, and variable-misidentification errors were 12% versus 24% (both $p<.001$). Consistent use of all five steps had the largest reported regression coefficient ($B=.45$, $SE=.09$, $p<.001$). Qualitative evidence indicated improved problem representation, confidence and cross-course transfer, alongside an initial time cost and resistance among some high-performing students. Conclusion: The evidence supports GUESS as a promising cross-disciplinary scaffold for well-structured quantitative problems, especially where novices need an explicit bridge from problem representation to execution. Replication should report group sizes, attrition, instrument reliability, ethics procedures and raw data so that effects can be independently verified.

Keywords: GUESS principle; STEM education; quantitative problem solving; metacognition; cognitive load; worked examples; transfer; University of Abuja

1. Introduction

Quantitative problem solving is not reducible to algebraic execution. A learner must first determine what information matters, represent the target, select a relation that is justified by the domain, substitute values without losing units or signs, execute the mathematics, and decide whether the answer is plausible. Novices frequently compress these decisions into a premature search for a formula. That shortcut can produce calculation activity without an adequate problem model.

The GUESS principle makes five of these decisions explicit: Given, Unknown, Equation, Substitute and Solve. Its educational premise is not that all STEM problems have a single linear pathway, but that an externalized routine can reduce avoidable representation errors while learners are developing domain-specific expertise. This positioning is consistent with research on scaffolding, worked examples, active problem solving and metacognitive monitoring. Recent evidence shows that active problem solving can improve test performance and reduce cognitive load across practice, while stepwise retrieval prompts embedded in worked examples can improve delayed problem-solving performance. These findings support the value of explicit, sequenced activity without implying that any one sequence should become a rigid algorithm (van den Broek et al., 2025; Zeitlhofer & Zumbach, 2026).

The present study is especially relevant because it evaluates one routine across chemistry, calculus, engineering mechanics and physics rather than treating problem solving as a purely discipline-specific skill. The central question is whether a common representation-and-execution scaffold can improve performance without erasing the conceptual knowledge unique to each field.

1.1 Research questions

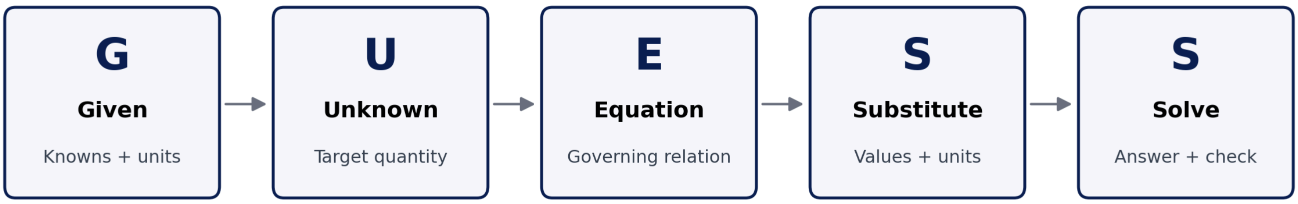
• Does instruction using the GUESS principle produce larger gains in calculation-based problem-solving performance than traditional discipline-specific instruction?

• Are any observed effects consistent across chemistry, calculus, engineering mechanics and physics?

• Which error patterns and learner experiences help explain how the framework operates in practice?

2. Conceptual Basis of the GUESS Principle

The five steps can be interpreted as a compact problem-representation scaffold. Given and Unknown force an explicit inventory of known quantities, constraints and the target. Equation requires a conceptual choice: the learner must identify a relation whose variables and assumptions connect the problem state to the target. Substitute delays numerical insertion until that relation is explicit. Solve then brings together calculation, units and a reasonableness check. The framework therefore separates representation from execution—an important distinction in cognitive-load accounts of novice problem solving.



Scaffold for well-structured quantitative tasks; domain knowledge, assumptions and verification remain essential.

Figure 1. GUESS as an explicit representation-to-execution scaffold. The final reasonableness/unit check is shown as part of "Solve" rather than as a sixth acronym step.

The model should be understood as a scaffold rather than a universal theory of problem solving. Open-ended design, modelling and ill-structured problems require iterative problem framing, assumption revision, evidence gathering and evaluation that may loop repeatedly across the five operations. GUESS is therefore most defensible for well-structured or moderately structured quantitative tasks and as an entry point to more sophisticated modelling practices.

3. Materials and Methods

3.1 Design and setting

The source study reports a sequential explanatory mixed-methods design at the University of Abuja, Nigeria, during the 2023–2024 academic year. Four undergraduate courses were included: General Chemistry I (n=78), Calculus I (n=65), Engineering Mechanics (n=42) and Introduction to Physics (n=56), giving N=241. Within each course, students were reported as randomly assigned to a treatment condition using GUESS or a control condition using traditional discipline-specific approaches. The source record does not provide the treatment/control allocation counts by course, attrition flow or allocation procedure; these omissions are retained as limitations rather than reconstructed.

Course	Reported enrolment	Instructional comparison
General Chemistry I	78	GUESS vs. traditional discipline-specific instruction
Calculus I	65	GUESS vs. traditional discipline-specific instruction
Engineering Mechanics	42	GUESS vs. traditional discipline-specific instruction
Introduction to Physics	56	GUESS vs. traditional discipline-specific instruction
Total	241	Four-course mixed-methods evaluation

3.2 Intervention and measures

Treatment instructors were reported to teach the five-step routine explicitly through worked examples, guided practice and independent problem sets. Controls continued with the existing discipline-specific problem-solving methods. Pre- and post-tests contained multi-step calculation problems. Responses were scored with a standardized rubric covering problem representation, strategy selection, implementation correctness, answer verification and overall solution efficiency. Structured error analysis tracked recurrent conceptual and procedural mistakes.

Qualitative evidence comprised semi-structured interviews with a stratified random sample of 40 students (10 per course, split across conditions), three classroom observations per course, instructor reflective journals, and think-aloud protocols with 20 students solving unfamiliar problems. These sources were thematically coded and compared to illuminate mechanisms behind the quantitative patterns.

3.3 Analysis and reporting boundary

The source manuscript reports comparative tests, a two-way ANOVA, regression modelling and error-frequency comparisons. This Version-of-Record-style article reproduces the reported aggregate results and strengthens their interpretation; it does not recreate statistics from raw data because no row-level dataset accompanied the supplied manuscript. Accordingly, reported coefficients and p values are treated as author-reported study results rather than independently reproduced estimates.

4. Results

4.1 Pre–post performance

Table 1. Reported pre-test and post-test performance by discipline and instructional group.

Discipline	Group	Pre-test mean (SD)	Post-test mean (SD)	Gain	Reported p
Chemistry	GUESS	68.3 (12.4)	84.7 (9.8)	+16.4	<.001
Chemistry	Control	69.1 (11.8)	75.3 (10.5)	+6.2	.023
Calculus	GUESS	71.5 (13.2)	86.2 (8.7)	+14.7	<.001
Calculus	Control	70.8 (12.5)	77.4 (11.2)	+6.6	.018
Engineering	GUESS	65.7 (14.1)	83.9 (10.3)	+18.2	<.001
Engineering	Control	66.2 (13.8)	72.6 (12.7)	+6.4	.031
Physics	GUESS	63.9 (15.2)	81.4 (11.6)	+17.5	<.001
Physics	Control	64.5 (14.9)	73.1 (13.8)	+8.6	.012

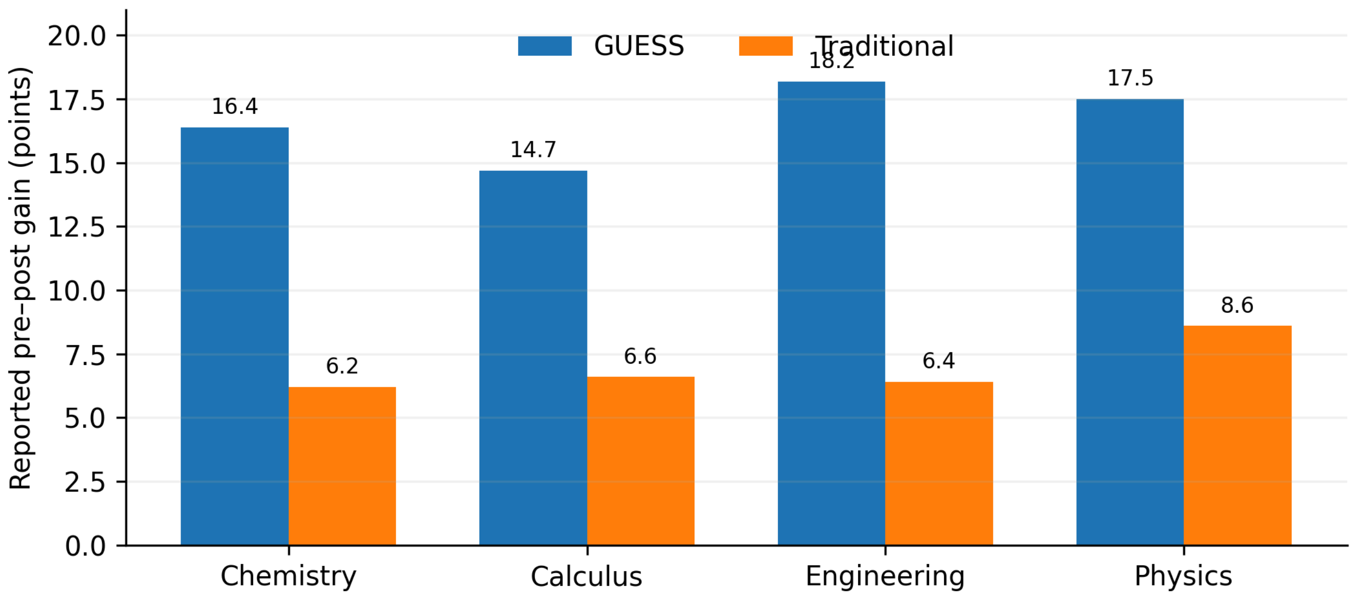


Figure 2. Reported course-level pre–post mean gains. The chart visualizes the source table; it does not imply equal treatment/control group sizes, which were not reported.

Across the four courses, the reported GUESS gains ranged from 14.7 to 18.2 points; control gains ranged from 6.2 to 8.6 points. As a descriptive summary across courses—not a participant-weighted effect—the mean reported gain was 16.7 points for GUESS and 7.0 points for controls. The reported two-way ANOVA indicated a main effect of teaching approach, $F(1,233)=42.68$, $p<.001$, partial $\eta^2=.155$. The discipline main effect was not significant, $F(3,233)=1.87$, $p=.136$, and the approach×discipline interaction was not significant, $F(3,233)=0.94$, $p=.422$. Within the source analysis, this pattern was interpreted as evidence that the instructional advantage was not confined to a single subject.

4.2 Error patterns and predictors

Table 2. Reported conceptual error frequencies.

Error type	GUESS	Control	Reported p	Interpretive link
Equation selection	16%	36%	<.001	Equation step: explicit relation selection before substitution
Variable misidentification	12%	24%	<.001	Given/Unknown steps: clearer problem representation

Table 3. Reported regression coefficients for factors associated with improvement.

Factor	B	SE	Reported p
Prior mathematics achievement	.32	.08	<.001
Problem complexity	.28	.07	<.001
Consistent application of all five GUESS steps	.45	.09	<.001
Student self-efficacy	.24	.07	.002
Instructor experience with the method	.18	.08	.025

Consistent application of all five steps had the largest reported coefficient ($B=.45$). This supports an interpretation in which the framework functions as a coordinated sequence rather than as isolated reminders. It does not, however, establish that step consistency is exogenous: more capable, motivated or better-supported students may also be more likely to apply the full routine. The coefficient is therefore best read as an association within the reported model.

4.3 Qualitative findings

Thematic evidence converged on four implementation themes. First, students described more deliberate initial representation of problems rather than immediate formula search. Second, the routine appeared to reduce feelings of being overwhelmed by complex calculations. Third, some students reported using the same sequence across chemistry, physics and calculus, suggesting perceived transfer. Fourth, implementation was not cost-free: structured solutions initially took longer, some high-performing students resisted a prescribed routine, and instructors needed to adapt the approach for complex or open-ended tasks.

5. Discussion

The most defensible interpretation of the findings is that GUESS externalizes a sequence of decisions that novices often fail to coordinate. The largest reported reductions occurred in conceptual representation errors—equation selection and variable identification—rather than being framed merely as faster arithmetic. This matters because a wrong representation cannot usually be repaired by correct calculation.

The cross-course pattern is also important. Contemporary research on problem solving suggests that active problem-solving episodes support performance and metacognitive monitoring, while worked examples are most effective when their structure is clear and when learners are prompted to engage with solution steps rather than passively copy them (van den Broek et al., 2025; Zeitlhofer & Zumbach, 2026). GUESS can be located between these traditions: it is not a worked example by itself, but it provides a common syntax through which worked examples, guided practice and independent problems can be organized.

The absence of a significant approach×discipline interaction in the reported ANOVA supports cross-disciplinary promise, but it should not be inflated into proof of universal transfer. The study involved one institution, four courses and one academic year. Chemistry stoichiometry, mechanics, calculus and physics share quantitative structure, yet advanced engineering design, data-rich modelling and ill-structured scientific inquiry require iterative cycles beyond the five-step routine. A mature implementation should therefore teach GUESS as a scaffold that can be compressed, expanded or looped as expertise grows.

6. Implications for Teaching and Curriculum

Instructional phase	Recommended use of GUESS	Teacher action	Evidence to monitor
Modelling	Teacher demonstrates all five steps aloud	Explain why each quantity/equation is selected, not only what is written	Student explanation quality
Guided practice	Partially completed GUESS templates	Fade prompts as accuracy improves	Representation and equation-selection errors
Independent practice	Compact headings or mental checklist	Require units, assumptions and reasonableness checks	Transfer to unfamiliar problems
Assessment	Rubric awards representation and strategy marks	Separate conceptual from arithmetic errors	Error profiles and solution efficiency
Advanced/ill-structured tasks	Use GUESS inside iterative modelling cycles	Allow loops, alternative models and assumption revision	Quality of assumptions, models and validation

7. Limitations and Research Transparency

The study record has material reporting gaps. Treatment/control group sizes by discipline, attrition, randomization details, instrument reliability, exact test content, regression model specification and ethics/consent documentation were not supplied. Raw data were also unavailable for independent reanalysis. These omissions do not erase the reported pattern, but they limit reproducibility and the strength of causal claims. Future replications should preregister allocation and analysis, publish de-identified data and scoring rubrics where permissible, report reliability and inter-rater agreement, and include delayed-transfer measures rather than relying only on immediate post-tests.

8. Conclusion

Across four undergraduate STEM disciplines, the supplied study record reports larger performance gains, fewer problem-representation errors and consistent qualitative benefits for students taught with the GUESS principle. The strongest contribution is not the acronym itself but the instructional separation of problem representation, relation selection and numerical execution. Used flexibly, GUESS offers a plausible scaffold for novice quantitative reasoning. Its next research stage should move from promising single-institution evidence to transparent multi-site replication with complete reporting and delayed transfer tests.

Declarations

Declaration	Statement
Ethics and consent	Not stated in the supplied study record; no unsupported approval or consent details have been added.
Funding	No external grant funding is declared in the supplied manuscript.
Competing interests	The author declares no competing interests.
Data availability	No row-level dataset was supplied; reported aggregate statistics could not be independently reanalysed.
Author contributions	Chia Shiaondo Kenneth: GUESS concept, study/source manuscript, interpretation and authorship.
AI/editorial preparation	Substantive editorial reconstruction and formatting; no unsupported study results were added.

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